

1. A mixture of two solid elements with a mass of 1.52 g was treated with an excess of hydrochloric acid. A volume of 0.896 L of a gas was liberated in this process and 0.56 g of a residue remained which was undissolved in the excess of the acid.

In another experiment, 1.52 g of the same mixture were allowed to react with an excess of a 10 % sodium hydroxide solution. In this case 0.896 L of a gas were also evolved but 0.96 g of an undissolved residue remained.

In the third experiment, 1.52 g of the initial mixture were heated to a high temperature without access of the air. In this way a compound was formed which was totally soluble in hydrochloric acid and 0.448 L of an unknown gas were released. All the gas obtained was introduced into a one liter closed vessel filled with oxygen. After the reaction of the unknown gas with oxygen the pressure in the vessel decreased by approximately ten times ($T = \text{const}$).

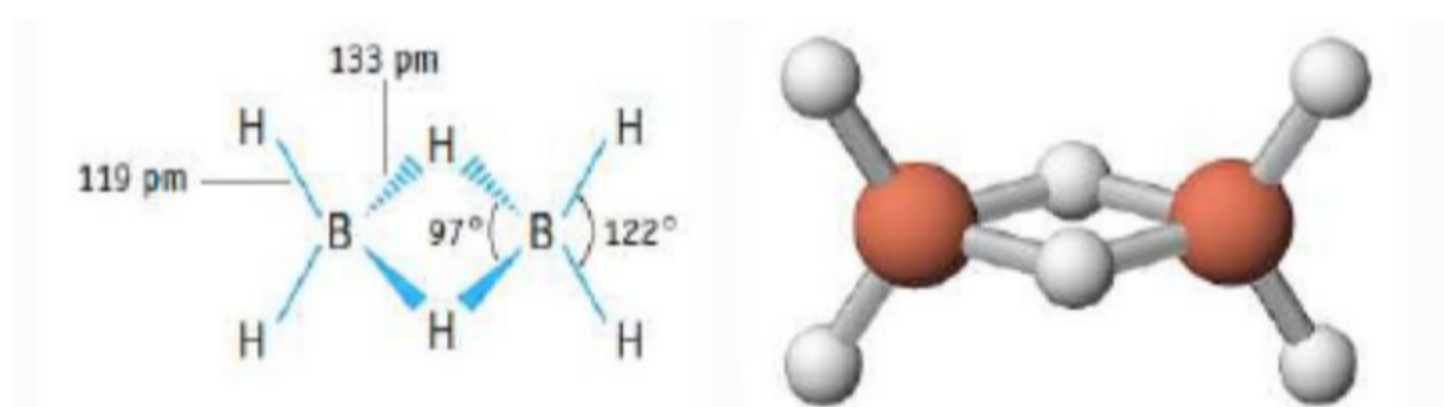
Problem: Write chemical equations for the above reactions and prove their correctness by calculations.

In solving the problem consider that the volumes of gases were measured at STP and round up the relative atomic masses to whole numbers.

2. This question is about Boron and its Compounds

A. Boron Hydrides and Other Boron Compounds

Boron hydride chemistry was first developed by Alfred Stock (1876 - 1946). More than 20 neutral molecular boron hydrides with the general formula B_xH_y have been characterized. The simplest boron hydride is B_2H_6 , diborane.



1. Using the data below derive the molecular formulae for two other members of this series of boron hydrides, **A** and **B**.

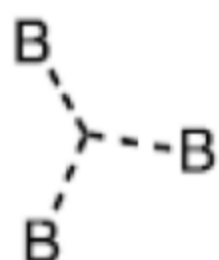
Substance	State (25 °C, 1 bar)	Mass Percent of Boron	Molar mass (g mol ⁻¹)
A	Liquid	83.1	65.1
B	Solid	88.5	122.2

William Lipscomb received the Nobel Prize in Chemistry in 1976 for “studies on the structures of boron hydrides illuminating the problems of chemical bonding.” Lipscomb recognized that, in all boron

hydrides, each B atom has a normal 2-electron bond to at least one H atom (B–H). However, additional bonds of several types occur, and he developed a scheme for describing the structure of a borane by giving it a styx number where:

s = number of B–H–B bridges in the molecule,

t = the number of 3-center BBB bonds in the molecule,



y = the number of two-center B–B bonds in the molecule,

x = the number of BH₂ groups in the molecule.

The *styx* number for B₂H₆ is 2002.

2. Propose a structure for tetraborane, B₄H₁₀, with a *styx* number of 4012.
3. A boron-based compound is composed of boron, carbon, chlorine, and oxygen. Spectral measurements indicate that the molecule has two types of B atoms, with tetrahedral and trigonal planar geometry, in a 1 : 3 ratio, respectively. These spectra are also consistent with a CO triple bond. The molecular formula of the compound is B₄CCl₆O. Suggest a structure for its molecule.

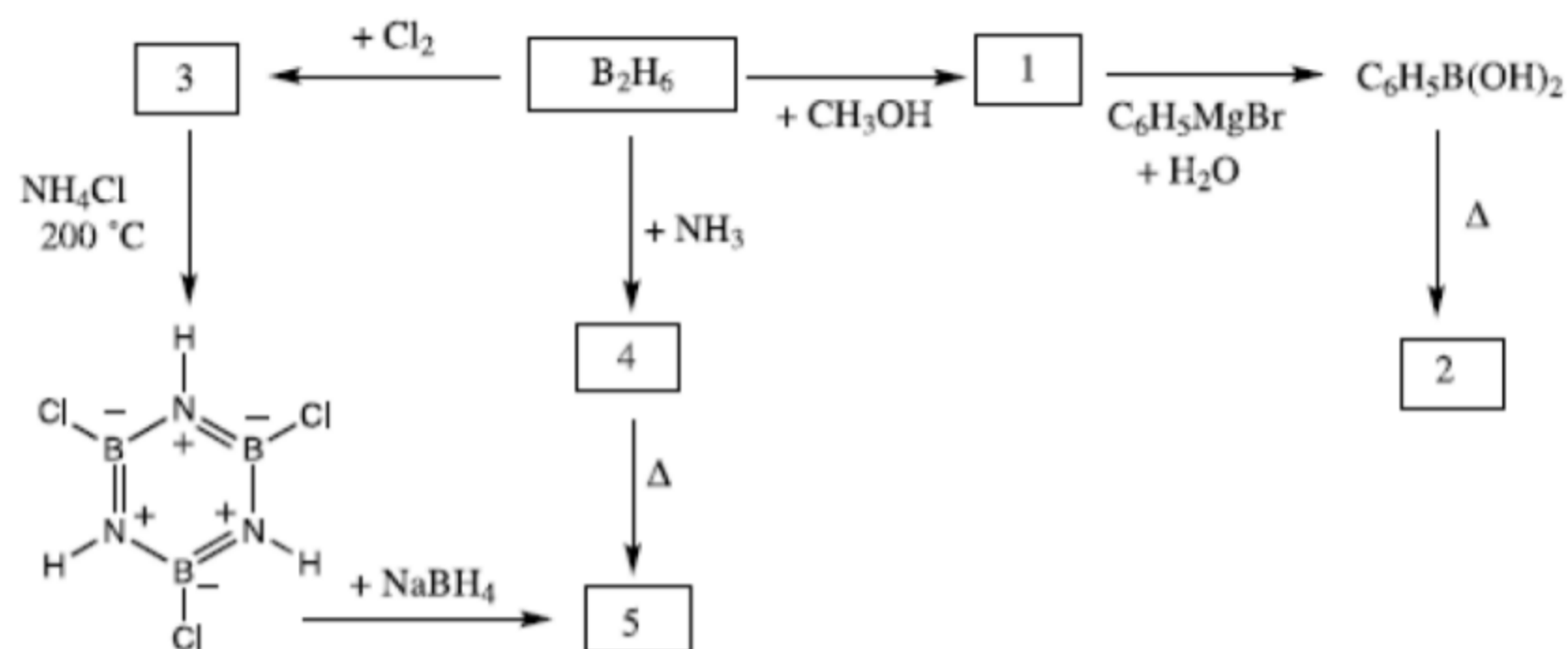
B. Thermochemistry of Boron Compounds

4. Estimate the B–B single bond dissociation enthalpy in B₂Cl₄(g) using the following information:

Bond	Bond dissociation enthalpy (kJ mol ^{–1})	Compound	$\Delta_f H^\circ$ (kJ mol ^{–1})
B–Cl	443	BCl ₃ (g)	–403
Cl–Cl	242	B ₂ Cl ₄ (g)	–489

C. Chemistry of Diborane

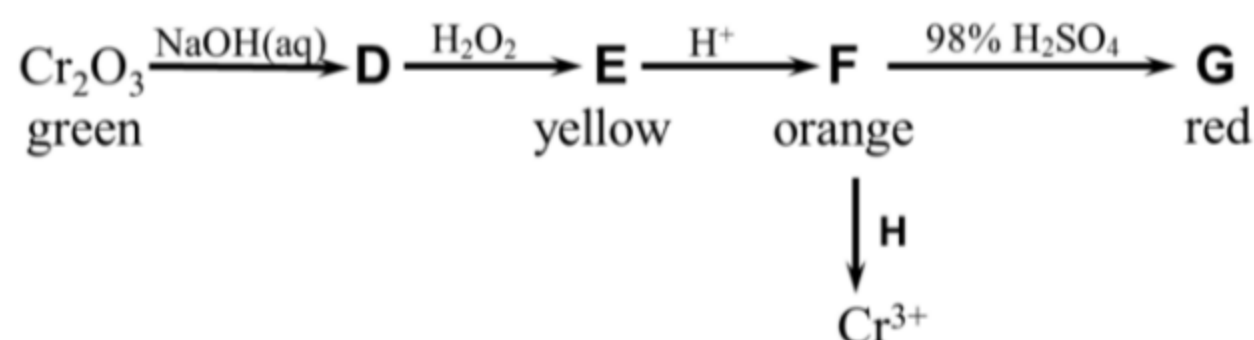
- 5 Give the structure for each numbered compound in the scheme below. Each numbered compound is a boron-containing compound.



NOTES:

- The boiling point of compound 5 is $55\text{ }^\circ C$.
- Excess reagents used in all reactions.
- The freezing point depression for 0.312 g of compound 2 in 25.0 g of benzene, is $0.205\text{ }^\circ C$. The freezing point depression constant for benzene is $5.12\text{ kg K mol}^{-1}$.

3. Besides doping in black ceramic glaze, chromium-containing pigments are used in painting and printing due to the abundant color originating from chromium with different oxidation states, such as +2, +3 and +6. The pigment, chrome green (Cr_2O_3) can be converted into a series of other compounds (**D-G**) in the following process, where **E**, **F**, **G** are yellow, orange and red respectively.



3-1: Write the chemical formula of **E**.

3-2: Write the reaction equation of **F** $\rightarrow$ **G**.

3-3: Choose the agent which could be **H**. (A) $FeSO_4$ (B) $FeCl_3$ (C) $ZnSO_4$ (D) $CuSO_4$

The variable valence of chromium is not only important for the production of pigments, but also useful for catalysis. For example, a typical Phillips catalyst for the polymerization of ethylene consists of chromium oxide species grafted onto a porous support material such as amorphous silica. The tetra-coordinated Cr(VI) species is the core in the pre-catalyst (**1**), which is proposed to be reduced quickly into a six-coordinated Cr(II) species (**2**) by the ethylene molecules at the beginning. **2** is proposed to further catalyze polymerization of ethylene molecules.

3-4: The reaction of **1** with ethylene can be traced by UV–vis spectra, which show the light absorption of **1** at 21500 cm^{-1} and the light absorption of **2** at 16700 cm^{-1} in the visible region, respectively. Choose the appropriate combination of colors of **1** and **2**.

- (A) orange and white, respectively
- (B) orange and blue, respectively
- (C) blue and orange, respectively
- (D) blue and white, respectively

3-5: Assuming that the Cr(II) ion in **2** is located in a regular octahedral crystal field with a splitting energy Δ_o of 16000 cm^{-1} , draw the configurations of d electrons of the Cr(II) ion in **2**, and calculate the crystal field stabilization energy (CFSE) for the Cr(II) ion in **2**. (Note: The pairing energy P for Cr(II) in **2** is 23500 cm^{-1})

3-6: Coordination compounds/ions exhibit paramagnetism when containing unpaired electrons, and the corresponding magnetic moment (μ) of transition metal cations is calculated by the function $\mu = \sqrt{n(n+2)}\mu_B$ where n is the number of unpaired electrons. Calculate the magnetic moment in term of μ_B for the Cr(II) ion in **2**.